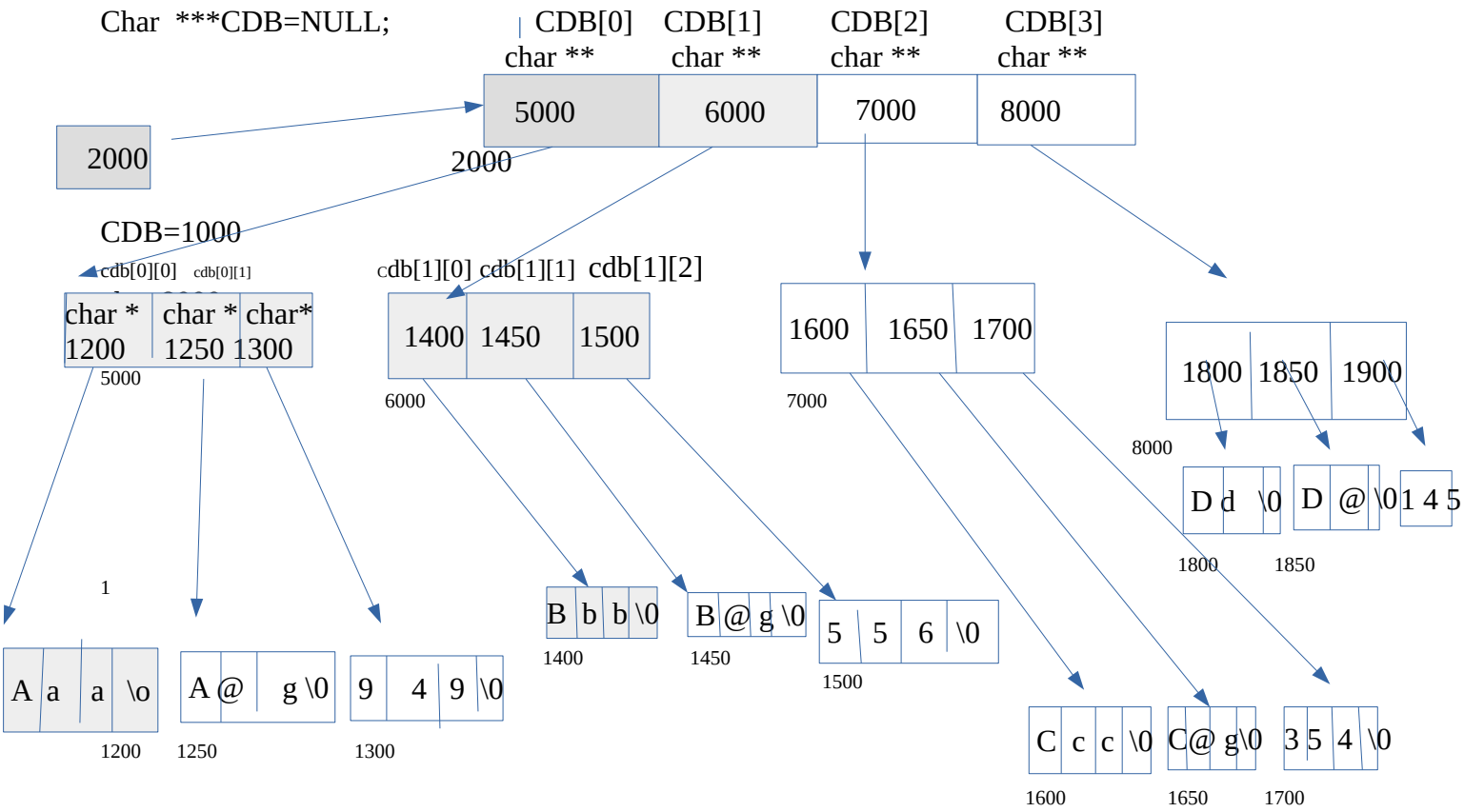


```
int arr[3][2][3]={ {1,2,3},{4,5,6}},
                  {{7,8,9},{10,11,12}} ,
                   { {13,14,15},{16,17,18} };
```

it has 3 ---->2D ARRAY
In each 2D ARRAY--->TWO 1D array
in each 1d array= 3 elements

****

3D array

1 time dereference=2d array
2 time dereference=1d array
3 times derefence=0th element

****

int (*p)[3][4];

sizeof(*p)= 8 bytes
sizeof(*p)=48 bytes//2d array
sizeof(**p)=16 bytes//1d array
sizeof(***p)=int //4 bytes

char *(q)[5][10];

sizoef(q)=8bytes
sizeof(*q)=50 bytes//2d array
sizeof(**q)=10 btes//1d array
sizeof(***q)=char //1 byte

.....

```
int main()
{
	int a[3][2][4];
	printf("%d\n",sizeof(a));//3d array//96 bytes
	printf("%d\n",sizeof(*a));//2d array//32 bytes
	printf("%d\n",sizeof(**a));//1d array //16 bytes
	printf("%d\n",sizeof(***a));///4 bytes

	char b[3][2][4];
	printf("%d\n",sizeof(b));//3d array///24 bytes
	printf("%d\n",sizeof(*b));//2d array//8 bytes
	printf("%d\n",sizeof(**b));///4 bytes
	printf("%d\n",sizeof(***b));///1 byte


}
```

Contact Database

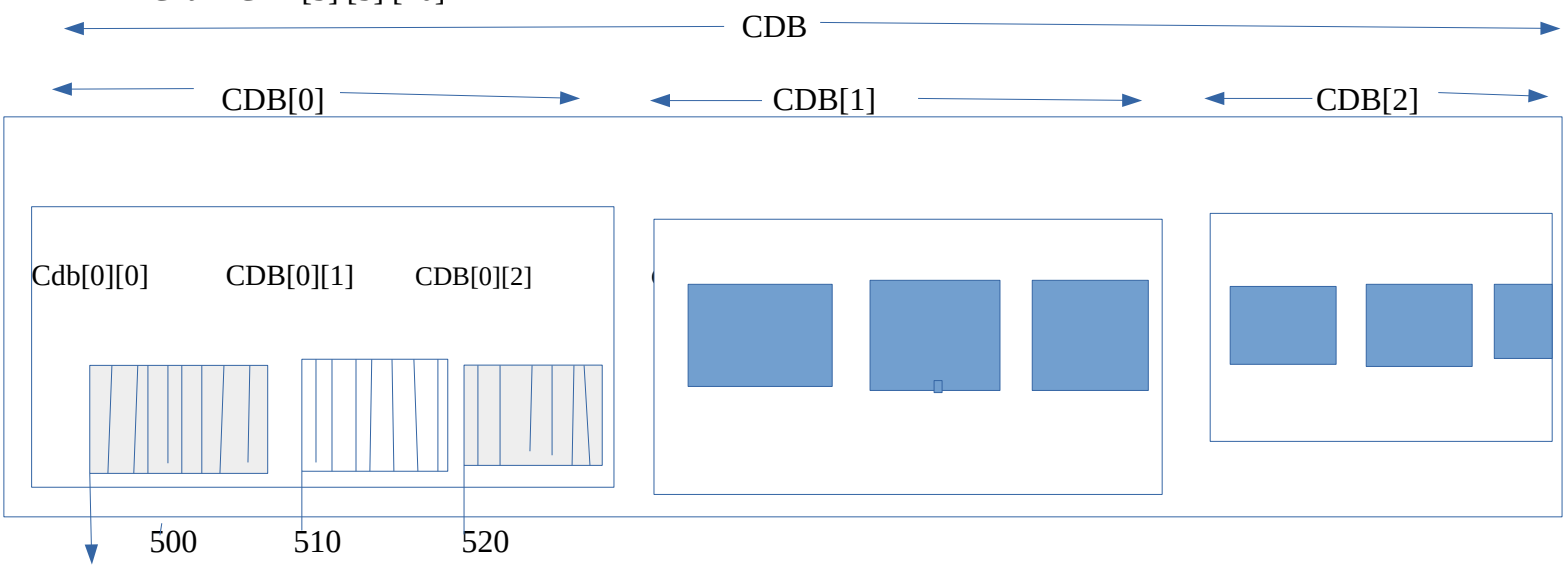
Suppose if we need to create a record which consists of information like Name ,Email Id ,conatct number of N (say as 10) number of students.

So each student should contain 3 fields or columns which include

1.Name
2.Email-ID
3.contact number

Let us create a record of 3 students which consists of Name, email id and mobile number.

Char CDB[3] [3] [10]



char CDB[8][3][10]

	CDB[i][0]	CDB[i][1]	CDB[i][2]
C DB[0]	AAA	a @gmail.com	1234...
CDB[1]	bbbb	b@gmail.com	3455...
CDB[2]	ccccc	c@gmail.com	56677..
CDB[3]	ddddd	d@gmail.com	3344..
CDB[4]	eeeee	e@gmail.com	455656..
CDB[5]	ffffff	f@gmail.com	4455...
CDB[6]	ggggg	g@gmail.com	3455..
CDB[7]	hhhhh	h@gmail.com	4455

Deleting IN conatct Database

char CDB[8][3][10]

	CDB[i][0]	CDB[i][1]	CDB[i][2]	
C DB[0]	AAA	a @gmail.com	1234...	CDB
CDB[1]	bbbb	b@gmail.com	3455...	CDB+1
CDB[2]	ccccc	c@gmail.com	56677..	CDB+2
CDB[3]	ddddd	d@gmail.com	3344..	CDB+3
CDB[4]	eeeee	e@gmail.com	455656..	CDB+4
CDB[5]	ffffff	f@gmail.com	4455...	CDB+5
CDB[6]	ggggg	g@gmail.com	3455..	CDB+6
CDB[7]	hhhhh	h@gmail.com	4455	CDB+7

NAME =ddddd //index=3

```
void del(char (*CDB)[3][20])
{       int i;
        char name[20];
        printf("enter the name to be deleted\n");
        scanf("%s",name);
        for(i=0;i<cnt;i++)
        {
                if(strcmp(name,CDB[i][0])==0)
                {
            memmove(CDB+i,CDB+i+1,(cnt-i-1)*sizeof(*CDB));/////8-3-1*sizeof(*CDB)
            cnt--;
            return ;

                }
        }
        printf("%s is not found\n",name);
}
```

Contact database using triple pointer

```
char ***CDB=NULL;
CDB=insert(CDB);


insert(char ***CDB)
{
        CDB=realloc(CDB,(cnt+1)*sizeof(char **));
        CDB[cnt]=calloc(3,sizeof(char *));
        printf("enter the name\n");
        CDB[cnt][0]=getString();
        printf("enter the mailid\n");
        CDB[cnt][1]=getString();
        printf("enter the contact\n");
        CDB[cnt][2]=getString();
        cnt++;
       return CDB;
}
.....................................................................................
char * getString()
{
        char *q=NULL;
        int i=0;
        __fpurge(stdin);
        do
        {
                q=realloc(q,i+1);
                q[i]=getchar();
        }while(q[i++]!='\n');
        q[i-1]='\0';
        return q;
}
```